



江苏省苏州中学 国际部
Suzhou High School - International Division

SZZX-ID Bridge Program Physics

free-body diagram

Class Outline

Title of Unit	Space, time and motion
Title of Lesson	Free body diagram
Starter Activity	describe a force and its impact (think pair and share)
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	1. Demonstration: use FBD to show force 2. Inquiry: FBD worksheet
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	1. Group work to learn types of forces and present. Teacher to support the flip-teaching
	International Mindedness
	FBD is a tool used for physics communication that does not require much language skills
Lesson Objectives	By the end of the lesson, students will be able to: 1. Use vector to represent force on a free body 2. Resolve a force into x and y component
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	1. Communication – use symbol and sign (FBD)
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Communicators
Teaching Activities	1. FBD worksheet 2. Resolve a force 3. Kahoot - FBD
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	FBD diagram is a tool for you to visualize force. Draw a FBD whenever you can.

Vocabulary



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- Force 力
- Diagram 图示
- Components 组成部分; (矢量的) 分量
- Horizontal 水平的
- Vertical 竖直的
- Normal 垂直
- Perpendicular 垂直
- Magnitude 大小

Representing force: free-body diagram



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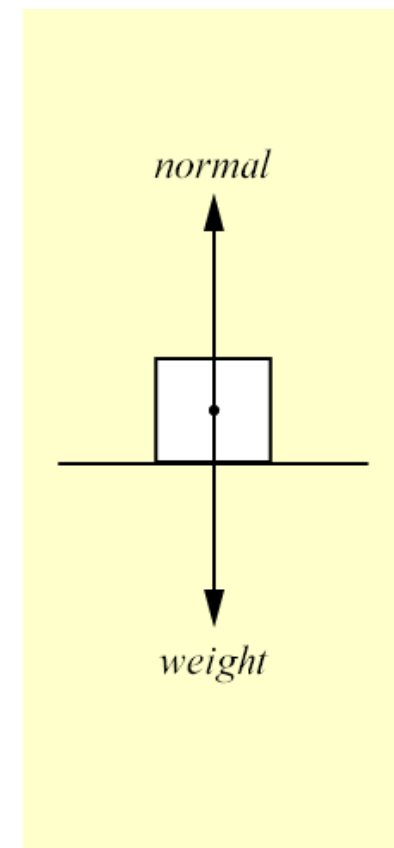
- A free-body diagram illustrates the relative magnitude and direction of all forces acting upon an object. The object must be isolated and “free” of its surroundings.
- Show the magnitude and direction of all forces acting on an object
- Magnitude is shown by the length of the vector arrows
- The object is isolated from the environment, free of its surroundings and of any other bodies it may be in contact with
- Treat the object as a point, so that all the forces act upon that point

free-body diagram example



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Free body diagram:
A book lying on a level table

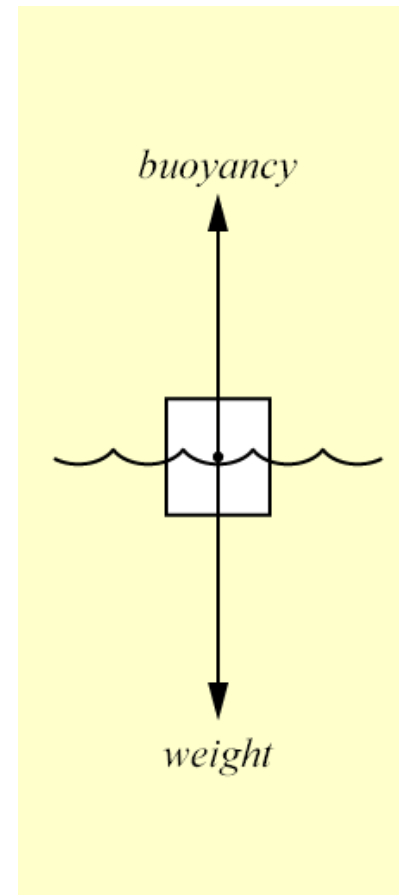


free-body diagram example



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Free body diagram:
A person floating in still water

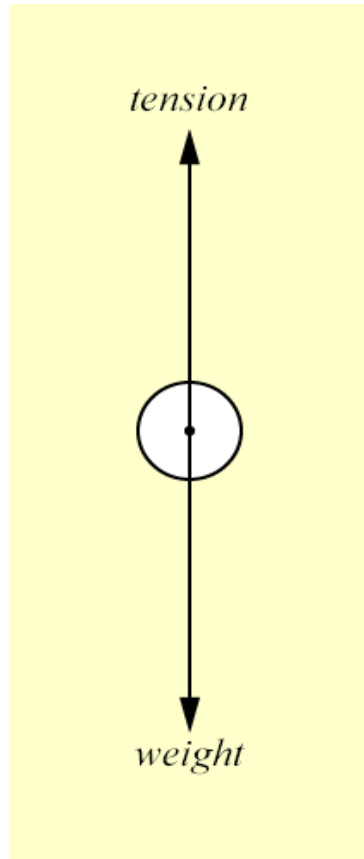


free-body diagram example



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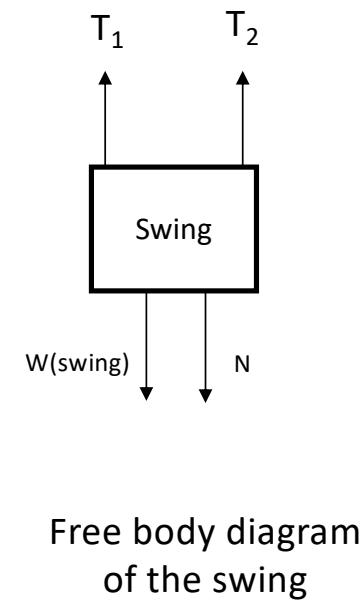
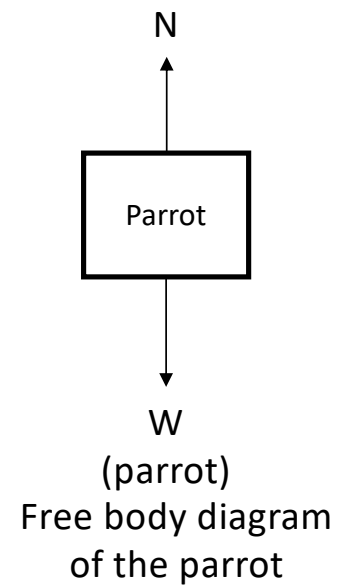
Free body diagram:
A wrecking ball hanging vertically from a cable



free-body diagram example



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Representing force: free-body diagram



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Let's try to draw the FBD for the following object:

1. A book is at rest on a tabletop.
2. An egg falls from a nest in a tree.
3. A cube stays on the slope.

Equilibrium condition



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- **Objects that are either at rest or moving with constant velocity are said to be in equilibrium ($a=0$)**
- **For any object in equilibrium, the net force acting on it (vector sum of all forces) is 0**
- A book lying still on the desk is in equilibrium
- Net force acting on the book is hence 0. $N = W$
- Qn: if the box is pushed, and moves on the ground with constant velocity, can you draw the FBD for the box?



Components of a vector



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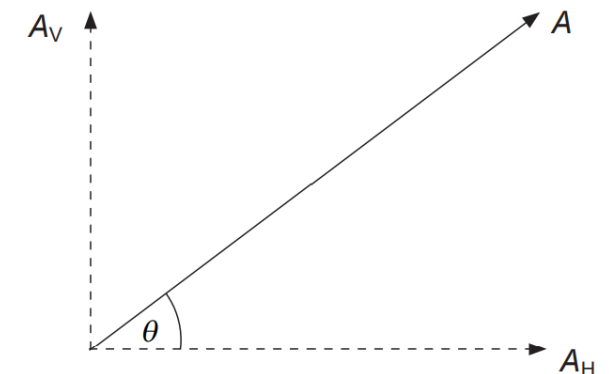
- To resolve a vector into components along x and y directions

The x- and y-components of A are called A_x and A_y . They are given by:

$$A_x = A \cos \theta$$

$$A_y = A \sin \theta$$

where A is the magnitude of the vector and the axis



$$A_H = A \cos \theta$$

$$A_V = A \sin \theta$$

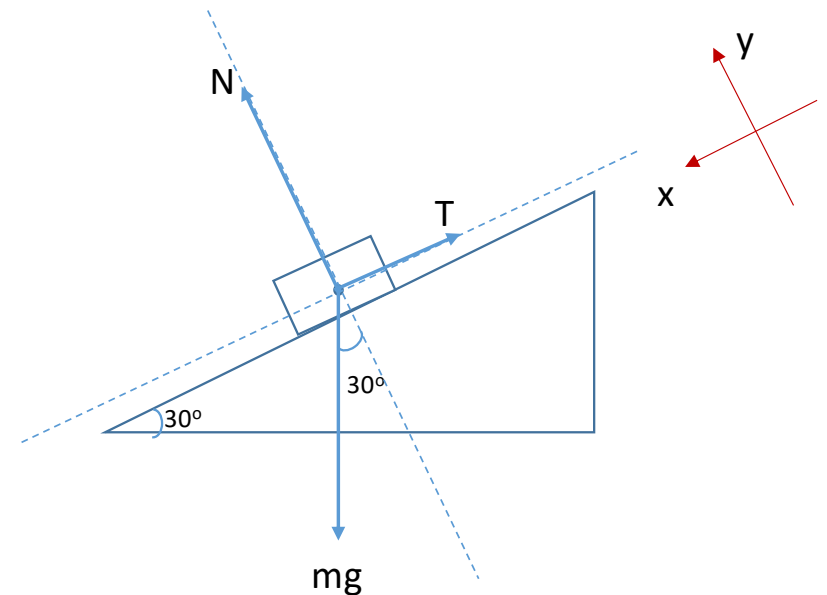
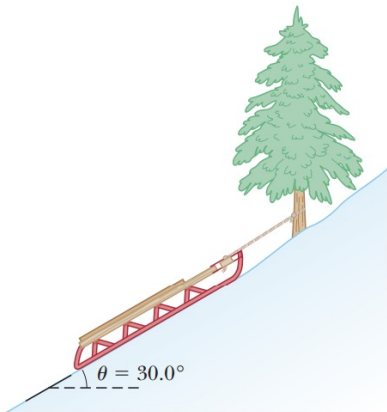
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$$\sum \vec{F} = 0 \quad \text{or} \quad \sum F_x = 0 \quad \text{and} \quad \sum F_y = 0$$



How would you draw the free-body diagram of the sled? Slope is smooth so ignore friction for now.

Equilibrium condition



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If we know the mass of the sled m , can we work out the magnitude of each force?

The sled is not moving $\Rightarrow \sum \vec{F} = 0$

- Define the positive x direction downward along the slope
- Define the positive y direction perpendicular to the slope, pointing outwards

$$\sum F_x = 0 \quad \text{and} \quad \sum F_y = 0$$

- Along the x direction: $mg \sin 30^\circ - T = 0$ (equation 1)
- Along the y direction: $mg - G \cos 30^\circ = 0$ (equation 2)

As $G = mg$, substitute G into equation 1 and 2:

$$\Rightarrow T = \frac{1}{2}mg \approx 4.9 \text{ m}$$

$$N = \frac{\sqrt{3}}{2}mg \approx 8.5 \text{ m}$$

